Frustration Markers in Text-Based Tutoring Dialogues: A Lexicon and Validation Study

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Abstract

We present a lexicon of 214 textual markers of frustration in learner turns of text-based tutoring dialogues, grouped into six categories. Markers were derived from 1,800 annotated dialogues and validated against in-session self-reports from 240 learners. A simple marker count correlated with self-reported frustration at r = 0.47, comparable to a fine-tuned classifier (r = 0.52) and far more interpretable. Repetition of a prior question with added negation was the single strongest marker.

1 Introduction

Frustration in face-to-face tutoring shows in posture and voice. In text, learners have fewer channels, but they still signal: they shorten replies, repeat questions, and say they are giving up. We catalogue these signals.

2 Building the lexicon

Two annotators marked spans in 1,800 dialogues from an online algebra tutor where learners had rated their frustration every five minutes. Candidate markers were kept if they appeared in at least 15 dialogues and were more frequent in high-frustration windows.

Category	Markers	Example
Repetition with negation	31	"no I mean why doesn't it ..."
Terse replies	18	"ok", "k", "sure" after long explanations
Explicit giving up	27	"forget it", "I'll just guess"
Capitalization and punctuation	22	"WHY", "???"
Self-deprecation	41	"I'm so bad at this"
Meta-complaints about the tutor	75	"you already said that"

3 Validation

On 240 new learners, the count of markers in a five-minute window correlated with self-reported frustration at r = 0.47. A fine-tuned transformer reached r = 0.52. Combining both reached r = 0.55. The strongest single marker was repetition with negation (odds ratio 3.9 for high frustration).

4 Discussion

Terse replies were ambiguous: some learners were frustrated, others were simply efficient. Markers should be interpreted relative to a learner's own baseline. Self-deprecation was frequent but weakly predictive; learners used it jokingly.

5 Limitations

One domain (algebra), English only, and self-reports that may themselves be affected by the act of reporting.

References

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[2] Russell, J. A. A circumplex model of affect. Journal of Personality and Social Psychology 39, 6 (1980), 1161-1178.